Martine: Benchmarking Multi-View 3D Surface Reconstruction Across Viewpoint Coverage, Resolution, and Lighting

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Abstract

Controlled acquisition setups are widely used for high-accuracy multi-view 3D surface reconstruction, as they allow sensor configuration, image resolution, viewpoint distribution, and illumination to be specified. Yet benchmarking remains difficult, since reconstruction accuracy depends jointly on the reconstruction paradigm, object properties, and acquisition regime. Existing benchmarks often restrict viewpoint coverage, resolution, or illumination, which can bias comparisons toward methods suited to the chosen setting. Others lack accurate 3D ground truth or quantified image-to-geometry alignment, limiting their use for metric surface evaluation. We introduce Martine, a benchmark for controlled 3D object surface reconstruction. Martine comprises 28 objects, including transparent materials and fine structures, captured at 9568×6376 pixels from 84 views over a spherical cap under both single-light and multi-light regimes. It provides object-independent camera calibration, accurate 3D ground-truth scans, calibration and image-to-geometry accuracy quantified on test images, and 3D property masks for analysing errors by material properties, visibility, curvature and fine structures. Experiments across photogrammetry, classical multi-view stereo, neural implicit surfaces, Gaussian-splatting-based reconstruction, and multi-view photometric stereo show that Martine enables regime-aware evaluation with explicit control over reference-data accuracy. The benchmark resolves sub-millimetric reconstruction differences, with a best observed mean one-sided Chamfer distance of 0.063 mm over the evaluated visible surface.

1 Introduction

3D surface reconstruction is a central problem in computer vision, with applications in cultural heritage preservation, industrial inspection, robotics, augmented reality, and digital content creation. Across these applications, the objective is often not only to recover a visually plausible shape, but to estimate surface geometry with sufficient accuracy for measurement, analysis, or downstream processing. Multi-view RGB surface reconstruction has achieved increasingly accurate results, in particular through advances in photogrammetry software and classical multi-view stereo (MVS) [1–5], neural implicit surface methods [6, 7], and recent multi-view multi-light variants [8, 9]. While 3D reconstruction can be performed in unconstrained environments [10–14], controlled acquisition setups provide important advantages when high accuracy is required. Such setups allow the acquisition

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Table 1: Comparison of object-scale datasets for controlled surface reconstruction. **360° Az.**, **Multi-Elev.**, **Dense Lights**, and **UHD** indicate full azimuthal coverage, multiple elevation levels, dense controlled illumination, and ultra-high-resolution imagery, respectively. **Image–GT Res.** indicates whether object-level residuals between evaluation images and the 3D reference geometry are reported, in addition to camera-calibration or scanner-accuracy statistics. **Challenge Eval.** indicates whether official masks or groups are provided to analyse errors by surface or material challenge. *n.r.* means not reported.

Dataset	360° Az.	Multi-Elev.	Dense Lights	UHD	Image–GT Res.	Challenge Eval.
DTU [15]	×	✓	×	×	*n.r.*	×
Luces-MV [18]	✓	×	✓	×	*n.r.*	×
DiLiGenT-MV [19]	✓	×	✓	×	*n.r.*	×
Skoltech3D [16]	×	✓	×	✓	*n.r.*	×
OpenSubstance [20]	✓	✓	✓	×	*n.r.*	×
EvalMVX [21]	✓	×	✓	×	*n.r.*	×
Ours	✓	✓	✓	✓	✓	✓

conditions to be specified in terms of sensor selection and configuration, image resolution, viewpoint distribution and illumination.

Existing controlled benchmarks cover this acquisition space only partially. DTU [15] and Skoltech3D [16] provide dense but azimuth-limited coverage, rather than broad object-centric coverage for multi-view surface reconstruction. While both include several illumination conditions, the lighting variations are primarily used to evaluate multi-view single-light methods under different static illuminations, rather than to support multi-view photometric stereo, which requires dense illumination sampling. This limitation is quantitatively studied by [17], which evaluates the suitability of existing controlled datasets for photometric-stereo-based surface reconstruction.

Conversely, datasets designed for multi-view photometric stereo combine several viewpoints with multiple controlled illumination conditions, so that surface geometry can be inferred not only from multi-view consistency but also from shading variation. These datasets provide illumination conditions suitable for photometric reconstruction, but each covers the remaining benchmark dimensions only partially, such as viewpoint density, elevation coverage, image resolution, object count, or surface ground truth. DiLiGenT-MV [19] remains the standard benchmark for multi-view photometric stereo, but contains only five objects, each captured from 20 views at 512×612 resolution. More recent datasets increase resolution or object count, but remain limited in viewpoint elevation coverage: Luces-MV [18] contains 12 views at 2368×1552 resolution, and EvalMVX [21] contains 20 views at 2448×2048 resolution; all three acquire views at a single elevation. Such design choices can also bias cross-paradigm comparisons. EvalMVX [21], for example, compares single-light and multi-light reconstruction using 20 views over 360° azimuth at a single elevation, rather than a denser single-light regime such as the widely used DTU benchmark [15], which provides 49–64 views over approximately 0°–110° azimuth. OpenSubstance [20] substantially improves view and illumination coverage, with 270 viewpoints and more than 1000 lighting conditions, and is a valuable resource for novel-view synthesis, appearance modelling, and object relighting. However, for surface-reconstruction benchmarking, its image-to-geometry alignment relies on object-dependent registration between the ground-truth scan mesh and a COLMAP [22] reconstruction. This can be unreliable when the COLMAP reconstruction is incomplete or geometrically inaccurate (see Figure 1), for example for transparent objects, thin structures, or challenging materials; yet the resulting alignment errors are not systematically quantified or reported. A related concern is addressed by NAVI [14] for in-the-wild object reconstruction, where 3D scanner ground truth is combined with manually verified 2D-3D alignment to obtain accurate object alignments. In addition, OpenSubstance ground-truth scans are not publicly provided for all objects, which limits surface-reconstruction evaluation on challenging cases.

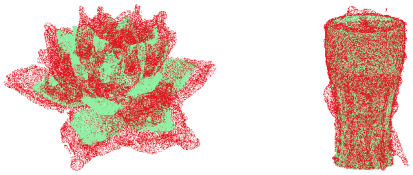


Figure 1: Examples of image–GT misalignment in OpenSubstance [20]. Inaccurate COLMAP reconstructions of an opaque flower (left) and a transparent glass (right) in red. Registered GT scan meshes in green.

Similarly, Luces-MV [18] does not provide object-to-camera reference alignment for its transparent objects.

A further issue is that the RGB input used by single-light reconstruction methods in multi-view photometric-stereo datasets is often synthesized from the multi-light stack, for example by taking the median [23, 24] or average [18] image over illuminations at each viewpoint. This aggregation has no explicit image-formation basis for single-light reconstruction and is likely motivated by the mitigation of specularities, since specular highlights are often localised and therefore appear only in a subset of the illumination stack. These results remain valuable, but they correspond to a hybrid setting, namely multi-light acquisition followed by single-light reconstruction. They therefore do not directly measure the performance of single-light reconstruction methods under a single-light acquisition regime. As a result, current benchmarks do not jointly provide broad object-centric viewpoint coverage, dense camera-mounted illumination, high-resolution imagery, accurately aligned 3D ground-truth scans, and acquisition regimes that allow single-light and multi-light methods to be compared under conditions designed for each paradigm.

Beyond viewpoint and illumination coverage, image resolution is another source of ambiguity in surface-reconstruction evaluation. This is particularly important for mature photogrammetry pipelines [2, 3], whose patch-based matching procedures can benefit directly from high-resolution imagery. While higher-resolution images provide more information, they do not consistently lead to lower reported reconstruction errors across existing benchmarks. As reported in [17], Luces-MV [18] and Skoltech3D [16] yield higher errors than DiLiGenT-MV [19], despite substantially higher image resolutions, 2368×1552 and up to 7296×5472, compared to 512×612. This discrepancy is difficult to interpret because resolution is confounded with dataset-specific differences in acquisition, objects, reference scans, and image-to-geometry alignment. It may indicate either that the reference data are not accurate enough for high-resolution evaluation, or that current implementations do not fully exploit high-resolution inputs. Distinguishing between these explanations requires benchmarks with high-resolution imagery, accurate 3D ground truth and quantified image-to-geometry alignment errors.

A further limitation concerns how reconstruction errors are reported. Most benchmarks report aggregate scores per object [15, 16, 18–21], although each object combines several geometric and optical properties. This makes it difficult to isolate failures due to visibility, curvature, fine structures, transparency, specularity, low texture, or other factors. Prior work has already shown that improvements on high-curvature or low-visibility regions can be poorly reflected by global Chamfer-distance or F-score averages [9, 17]. However, such analyses remain scattered across individual papers and are rarely integrated into public benchmark protocols.

To address these limitations, we introduce Martine, a controlled benchmark for 3D object surface reconstruction. Its main contributions are as follows.

- **A benchmark spanning acquisition regimes.** Martine comprises 28 objects placed on a motorised turntable and imaged with a high-end camera mounted on a robotic arm. For each object, we acquire 84 views over a spherical cap at 9568×6376 resolution under two lighting regimes. The single-light regime uses a reflecting cone and a softbox flash to approximate rotation-invariant ambient illumination around the turntable axis, while the multi-light regime uses 30 camera-mounted LED light sources activated sequentially. This design provides acquisition conditions suited to dense-view single-light reconstruction as well as multi-view multi-light reconstruction. The dataset includes challenging geometries and materials, including fine structures, transparent objects, low-texture surfaces, and specular or glossy regions.

- **Reference data with quantified accuracy.** Martine provides object-independent camera calibration, calibration accuracy measured on test views, accurate 3D ground-truth scans, and easy to interpret 2D–3D registration residuals on test images. These elements make it possible to evaluate surface reconstruction while explicitly accounting for reference-data accuracy.

- **Challenge-level performance analysis.** We provide evaluation code and 3D ground-truth masks to report reconstruction errors by optical properties, visibility, curvature and fine structures, rather than only by object identity.

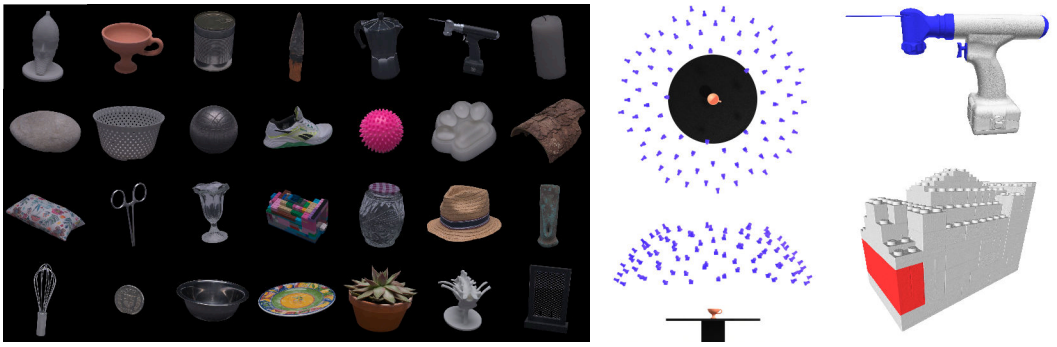
- **A first cross-paradigm evaluation.** We provide an initial evaluation of representative methods from photogrammetry, patch-based MVS, 3D Gaussian Splatting-based reconstruction, neural implicit surfaces, and multi-view photometric stereo.

Together, these contributions make Martine a benchmark for studying how acquisition regime, object properties, and reconstruction paradigm jointly affect surface-reconstruction accuracy.

2 Dataset and Acquisition Design

2.1 Overview

Objects. The object set is designed to cover a range of geometric and optical properties relevant to image-based surface reconstruction. It includes simple textured and matte objects, as well as more challenging cases with fine structures, deep concavities, transparent objects, low-texture surfaces, and highly specular materials. Objects are shown in Figure 2(a).



(a) (b) (c)

Figure 2: Dataset overview. (a) All objects in the proposed dataset, showing high geometric and optical diversity, including low-texture regions, fine structures, transparent and specular materials, and high-frequency geometric details. (b) Dense viewpoint sampling with wide angular coverage, obtained by coupling a robotic arm with a turntable. (c) Two examples of ground-truth meshes with manually curated challenge-level 3D masks highlighting specular regions (top, in blue) and transparent regions (bottom, in red) for challenge-level evaluation.

Viewpoints. The proposed viewpoints follow a Fibonacci-sphere distribution restricted to a spherical cap of 80 cm radius around the object. We acquire 84 views between 20° and 70° elevation, providing near-uniform sampling of the cap. Each view is captured at 9568×6376 resolution with a Sony Alpha 7RV camera. Viewpoints are shown in Figure 2(b).

Single-light setup. The single-light regime uses an ambient illumination setup designed to be approximately rotationally symmetric around the turntable axis. This yields nearly view-invariant lighting for the object volume above the turntable, and provides favourable conditions for standard multi-view reconstruction methods. In particular, it substantially reduces the view-dependent appearance changes that would occur if the object were rotated under a fixed asymmetric light source. See Figure 3(a). The rotational symmetry of the illumination was evaluated quantitatively by analysing intensity variations along the isophotes of a white 25 cm sphere placed at the centre of the turntable. The measured variation ranges from 0.5% to 2.5%. See details in the appendix (Section C).

Multi-light setup. The multi-light regime uses a rig of 30 LEDs rigidly mounted around the camera. This number of illumination conditions is consistent with the quantitative evaluation of the state-of-the-art photometric stereo method UniMS-PS [25], which reports no improvement from increasing the number of input lights beyond 30 in their setting. It also reflects a practical trade-off in our acquisition design, as each additional light source multiplies the number of images, while the dataset already provides high-resolution 16-bit imagery, dense viewpoints, and two acquisition regimes. This design provides dense illumination for multi-view multi-light reconstruction. See Figure 3(b).

Challenge-level masks. We provide 3D masks on the ground-truth geometry to support evaluation beyond object-level averages. The masks identify Lambertian or matte regions, low-texture regions, transparent parts, specular regions, high-curvature regions and fine structures. We also provide visibility levels, defined from the number of views observing each surface point. Low-visibility regions generally correspond to concavities or self-occluded areas, and can also be affected by inter-reflections. Examples of 3D masks are shown in Figure 2(c).

2.2 Acquisition Setup

View acquisition. The setup incorporates a Kuka LBR Med 14 industrial robotic arm and a Noxon 360° turntable with an 80 cm diameter. The combination of these two tools allows us to define viewpoints derived from a Fibonacci sphere. This specific trajectory is achieved physically through the ROS 2-managed synchronization of 10 turntable rotations and precise robotic arm movements (see Figure 3). The turntable is reset between the single-light and multi-light acquisition sequences. This step is intended to maintain consistency between the two capture modes.

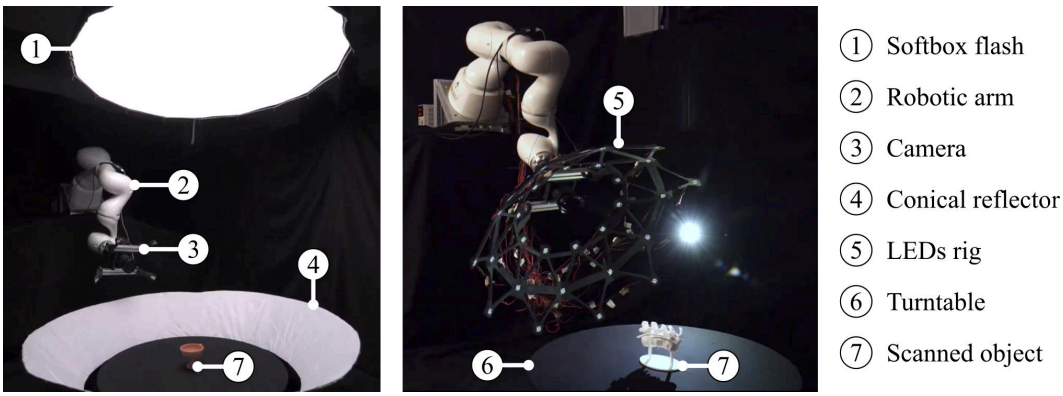


Figure 3: Acquisition setup. (a) Single-light acquisition regime, using an ambient illumination setup designed to be rotation-invariant around the turntable axis. (b) Multi-light acquisition regime, using a rig of 30 LEDs rigidly mounted around the camera. At each viewpoint, the LEDs are activated sequentially while the camera pose remains fixed.

Lighting setup. We acquire each object under two illumination regimes (Figure 3 left and right). In the single-light regime, the setup uses a 120 cm softbox flash together with a reflecting cone placed around the turntable to produce an approximately rotation-invariant ambient illumination around the object. In the multi-light regime, illumination is provided by a spherical-cap rig containing 30 Luxeon Rebel LEDs of 220 lm each. The 85 cm-diameter rig is rigidly mounted to the camera, ensuring a fixed relative configuration between the camera and the light sources across viewpoints. The LEDs cover a range of lighting directions around the optical axis while remaining at approximately the same distance from the object. Acquisition is performed in a darkroom, and the LEDs are activated sequentially following a one-light-at-a-time protocol. A black image per view point is also captured for the calibration.

2.3 Ground Truth Acquisition

We acquire reference geometry with an Artec Spider optical scanner [26], specified at 0.05 mm 3D point accuracy and 0.1 mm 3D resolution. As in high-accuracy object-reconstruction benchmarks such as Skoltech3D [16], specular and transparent objects are sprayed with Aesub Blue, a sublimating opacifying spray, to reduce optical artefacts during scanning.

The scans are acquired manually rather than with a fixed scanner trajectory. This choice is more time-consuming, but gives the operator direct control over the sensor path and allows coverage to be checked continuously during acquisition. It is particularly useful for objects with concavities, thin structures, or self-occluded regions, where a fixed scanner and turntable setup, may leave parts of the surface insufficiently observed. In total, we scanned 21 out of the 28 objects with the Artec Spider scanner. For objects with narrow concavities or structures that are difficult to observe with optical scanning, such as the plant, we complement the reference acquisition with computed tomography

scans acquired with a NAEOTOM Alpha CT scanner at a voxel size of 0.117 mm. The scans are then segmented using Materialise Mimics 26.0 [27].

This workflow follows a monitored pipeline in which each reference model is inspected manually. Among the 28 objects, two do not have ground-truth scans. The first is *pillow*, because its surface texture is partly interpreted by the Artec Spider as geometric variation. The second is *coin*, because neither scanning modality could capture its fine geometric details reliably.

2.4 Camera Calibration and Scan Alignment

This section describes the reference-data pipeline used to place all images and ground-truth scans in a common metric coordinate system. We first present the camera calibration and its validation, then describe the scan-alignment procedure and its validation.

Camera calibration. The camera-calibration procedure is designed to be object-agnostic while providing quantified accuracy. Object-agnostic calibration is important because camera poses should not depend on the geometry, texture, or material of the reconstructed object. Otherwise, pose errors could vary with object difficulty, making reconstruction errors harder to interpret.

Several strategies are possible. One option is to rely on the robot and turntable poses, but prior work [28] has shown that this can be insufficiently accurate at this image resolution. Another option, adopted by EvalMVX [21], is to place a calibration board below the object. This provides image features for pose estimation, but is not well suited to calibrated photometric stereo because the checkerboard can introduce inter-reflections and make object-contour extraction less robust during the subsequent alignment procedure. A related strategy, used in OpenSubstance [20], places markers on a vertical cylindrical support below the object, which likely reduces inter-reflections compared with a planar checkerboard. In contrast, our calibration uses only the textured black fabric covering the turntable, with the object and background excluded from feature extraction. In our evaluation, this provides the most accurate results while avoiding inter-reflections from markers.

Concretely, all images from each acquisition are calibrated jointly in a single Structure-from-Motion (SfM) run. This includes the single-light images, the multi-light images acquired with all LEDs active, and the opacifying-spray images when available. We use Segment Anything Model segmentation to generate turntable masks that restrict feature extraction to the black fabric region. SfM is performed with Meshroom [3], which builds on OpenMVG [29], using DSP-SIFT [30] features. Additional details on the calibration are provided in the appendix (Section A).

Calibration accuracy. We compare the proposed calibration procedure with three alternatives, some of which are representative of existing benchmarks. The first is COLMAP [4] applied to the same textured cloth. The second uses a ChArUco board [31], analogous to the ArUco-board calibration used with Metashape in EvalMVX [21]. The third uses an ArUco cylinder placed on the turntable base, analogous to the ARTag fiducial markers [32] used in OpenSubstance [20]. We use ArUco markers [31] instead of ARTag markers, as they have been reported to provide better performance in prior evaluations [33, 34]. Accuracy is evaluated on test images of an 18×18-corner checkerboard placed on the turntable, which is not used for calibration. Our SfM-based approach achieves the lowest reprojection error, with 0.542 px, corresponding to approximately 0.038 mm at the nominal object distance. This value should be interpreted as an upper bound on the calibration error, since the measured residual also includes checkerboard-corner localisation error. It is followed by the ArUco cylinder with 0.668 px, COLMAP with 0.809 px, and ChArUco with 1.086 px.

Scan alignment. We formulate scan alignment as a contour-based ICP problem in the image domain. The goal is to align the ground-truth mesh with the calibrated image sequence by matching projected mesh silhouettes to object contours extracted in the images. Since the ground-truth scans are expressed in metric units, while SfM camera poses are recovered up to an unknown scale, the alignment estimates a 7-DoF similarity transform consisting of rotation, translation, and uniform scale. The inverse transform is then applied to the camera poses, expressing them in the metric coordinate frame of the ground-truth scan and yielding a metrically consistent image-to-geometry alignment. Target contours are extracted from single-light images using Canny edge detection within a local search band defined from SAM object masks, with contour-extraction parameters optimised per object. For specular or transparent objects, we use images captured after applying the white sublimating opacifying spray, which increases the contrast between the object and the black turntable near the silhouette. The registration then optimises the similarity transform by

minimising the Euclidean distance between projected silhouette vertices of the ground-truth mesh and the corresponding target image contours. The mathematical formulation of the optimisation and the selection of the contour-extraction parameters are provided in the appendix (Section B).

Alignment accuracy. The alignment is validated on control views excluded from registration. For each object, we randomly select eight control views. At each iteration, the similarity transform is estimated from the remaining 83 views using the object-specific contour-extraction parameters, and the reprojection residual is measured on the excluded view. We report the average residual over the eight leave-one-control-view evaluations, yielding a pixel-level image-to-geometry error at the original sensor resolution. The mean LOO residual across all objects is 1.50 px, with a median of 1.36 px, and 24 out of 26 objects achieve a mean residual below 2 px, as shown in Table 2. We report reprojection residuals rather than silhouette IoU, as used in EvalMVX [21], because preliminary tests showed that IoU on downsampled masks can be insensitive to small contour misalignments. Additional details are provided in the appendix (Figure 9).

Table 2: Leave-one-out registration residuals (mean reprojection error in pixels, 8 control views per object).

rock	vase	hammer_head	candle	can	glass	petanque_ball	jam_pot
1.29	1.22	1.34	1.11	1.33	1.36	1.51	1.41

colander	wood	plate	paw_pad	moka_pot	lego	surgery_scissors	knife	spikes_ball
1.34	1.35	2.02	1.35	1.60	1.97	1.39	1.43	1.46

straw_hat	sneakers	surgery_saw	whisk	warrior_head	plant	grid	bowl	spine
2.09	2.52	1.28	1.33	1.34	1.32	1.49	1.32	1.43

Taxonomy of reconstruction challenges. Existing benchmarks typically report one aggregate score per object, even though a single object may contain several distinct surface properties, such as matte regions, low texture, high curvature, transparency, or specularity. As a result, an object-level score does not indicate which properties drive the reconstruction error. To make this analysis possible, we annotate each ground-truth mesh with per-vertex labels describing relevant geometric and optical properties. The seven property labels are defined as follows. *Lambertian* denotes matte regions whose appearance is approximately diffuse. *Specular* denotes regions with strong view-dependent reflectance, including glossy or metallic surfaces. *Transparent* denotes transmissive regions, for which image evidence can be affected by refraction, reflection, or background visibility. *Textureless* denotes regions with absent surface texture, which provide very limited local image correspondences. *Fine structure* denotes thin geometric elements with small cross-section, such as wires, blades and spikes. *High curvature* denotes vertices with large local maximum curvature, computed in our case over an empirically fixed neighbourhood of 2 mm and thresholded at -0.2 (convex) in our experiments. *Low visibility* denotes vertices observed in only a small number of calibrated views, thresholded at fewer than 3 views in our experiments. This property-level evaluation is particularly useful for regions that occupy only a small fraction of the object surface. Errors on such under-represented regions can be diluted in global metrics, such as mean F-score or Chamfer distance over the full object.

3 Experiments

3.1 Baselines

We evaluate representative methods from the reconstruction paradigms supported by the dataset. For photogrammetry, we use Meshroom [3], an open-source pipeline based on CMPMVS [35], and RealityScan [2], a widely used closed-source software. As a classical MVS baseline, we use COLMAP [4, 22]. We also evaluate PGSR [36] for 3D Gaussian Splatting-based surface extraction, and two neural implicit methods, NeuS2 [6] and Neuralangelo [7]. For multi-view photometric stereo, we evaluate SuperNormal [8] and RNb-NeuS2 [17], using the state-of-the-art photometric stereo method UniMS-PS [25] to estimate per-view normal maps from the multi-light images.

3.2 Evaluation Protocol

Image resolutions. Each method is evaluated at up to four image resolution levels: full resolution (9568×6376), half resolution (4784×3188), quarter resolution (2392×1594), and one-eighth resolution (1196×797). Only the two photogrammetry pipelines (Meshroom and RealityScan)

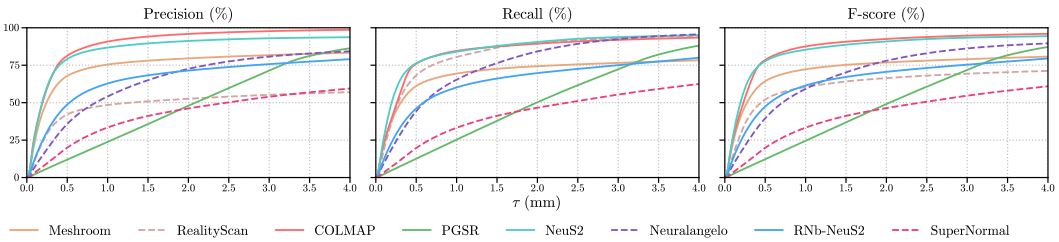


Figure 4: Global precision, recall, and F-score curves pooled across all evaluated objects. Each curve aggregates all point-to-surface distances before computing the metric, so larger objects contribute proportionally more.

process full-resolution images; all other methods are evaluated at lower resolutions due to GPU memory constraints.

Mesh cleanup. Reconstructed meshes are filtered by removing vertices not visible from any camera view. Unreliable regions of the ground-truth meshes (e.g., mounting bases) are excluded using per-object masks.

Evaluation scores. Surface accuracy is measured using precision, recall, and F-score curves for distance thresholds between 0 and 4 mm. Precision measures accuracy as the fraction of reconstructed points lying within a given distance of the reference surface. Recall measures completeness as the fraction of reference points lying within the same distance of the reconstruction. The F-score is the harmonic mean of precision and recall, summarising their trade-off.

3.3 Results

Figure 4 reports precision, recall, and F-score curves aggregated over all objects at quarter resolution. In this configuration, single-light methods generally achieve higher surface accuracy than the evaluated multi-light methods. COLMAP and NeuS2 show more consistent convergence across the object set, whereas RNb-NeuS2 and SuperNormal do not systematically benefit from the additional photometric information. This suggests that, for the present benchmark and implementations, dense multi-view constraints remain more reliable than the current integration of multi-light photometric cues. It also indicates that further work is needed to fully exploit photometric-stereo information in multi-view surface reconstruction.

Figure 5 studies the effect of reducing the number of views and the input resolution on the *vase* object. Classical single-light MVS shows a clear degradation under viewpoint sparsity, while NeuS2 and RNb-NeuS2 remain more stable in this example. RNb-NeuS2 performs best in the sparse-view setting, likely due to the normal cues provided by photometric stereo. Reducing image resolution has a stronger effect on COLMAP, whose patch-based matching benefits directly from high-resolution local image evidence. By contrast, NeuS2 and RNb-NeuS2 are more stable in the low-resolution regime, consistent with their pixel-wise optimisation through neural volume rendering.

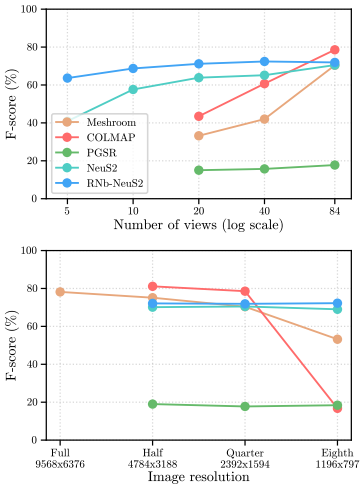


Figure 5: F-score at 1 mm on the *vase* object versus number of input views on a log scale (top) and image resolution (bottom). Missing points indicate failures due to excessive resolution or too few views.

Additional exemplar results are shown in Figure 6, which provides qualitative spatial error maps for representative objects and methods.

The property-level radar plot in Figure 7 shows the expected contrast between easier and harder surface properties. All methods perform best on Lambertian regions, while transparent regions lead to the largest errors. The performance drop is less pronounced on textureless regions. COLMAP performs best on Lambertian and high-curvature regions, while neural methods are the most competitive on specular regions, with NeuS2, and fine structures, with RNb-NeuS2. This is consistent with the

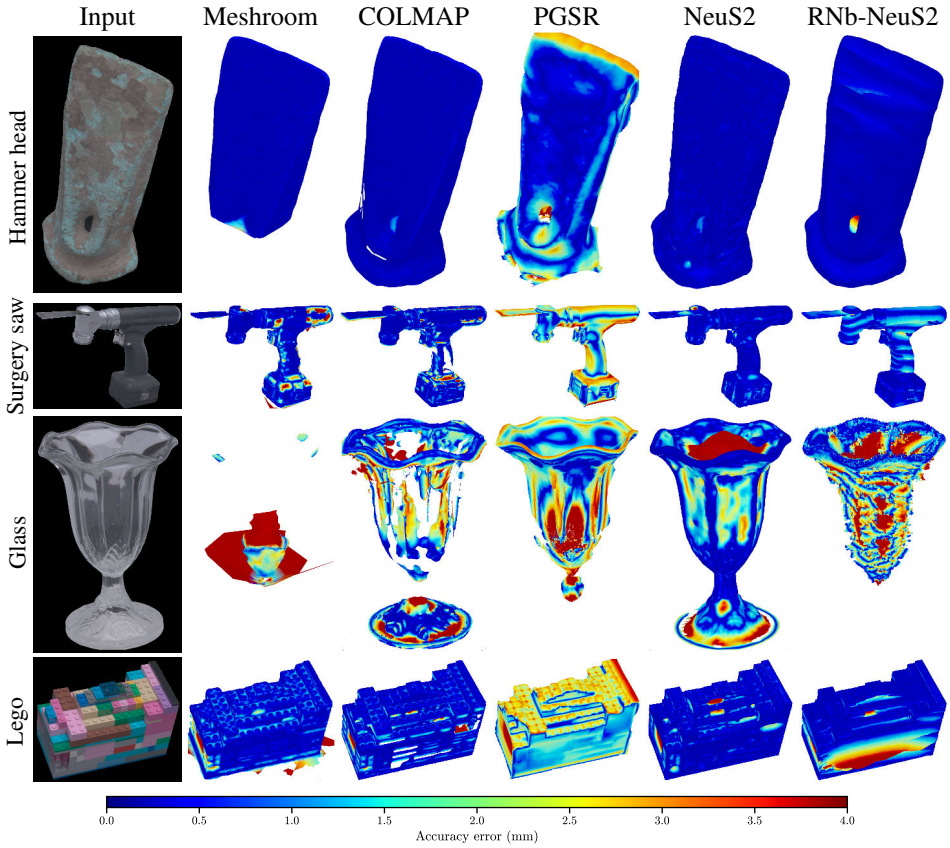


Figure 6: Qualitative comparison of accuracy error heatmaps (jet colormap, 0–4 mm) at quarter resolution. For reference, COLMAP achieves 0.063 mm on the hammer head at half resolution.

qualitative behaviour often reported for neural surface methods,
which tend to preserve thin or weakly supported structures better than classical fusion pipelines.

4 Conclusion and Future Work

In this work, we introduced Martine, a benchmark for object surface reconstruction in controlled environments designed to address limitations of existing datasets. Martine provides 28 objects, including challenging cases such as transparent materials and fine geometric structures, and controls acquisition parameters such as viewpoint distribution, image resolution, and illumination. This enables reconstruction paradigms to be compared on the same objects under well-defined acquisition regimes. The emphasis on reference-data accuracy, through object-independent calibration and quantified image-to-geometry residuals, supports a clearer interpretation of reconstruction errors. In addition, 3D property masks enable a more detailed analysis of errors across surface regions and object properties.

The current version of the dataset nevertheless has limitations. First, the property taxonomy is partly qualitative. Categories such as Lambertian, specular, textured, textureless, or fine structure would benefit from more formal definitions and objective annotation criteria. Several of these properties are continuous rather than discrete, and future evaluations should ideally report performance along such dimensions rather than only through hard labels. More generally, we hope that this benchmark will encourage the

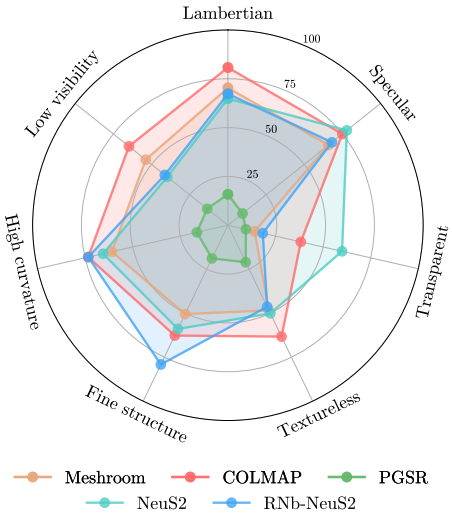


Figure 7: Radar chart of F-score at 0.5 mm per reconstruction challenge. Higher is better.

community to characterise reconstruction methods by the technical challenges they address, rather than only through aggregate rankings. We also hope it will promote more systematic reporting of ground-truth accuracy, camera-calibration accuracy, and image-to-geometry alignment errors for this type of benchmark.

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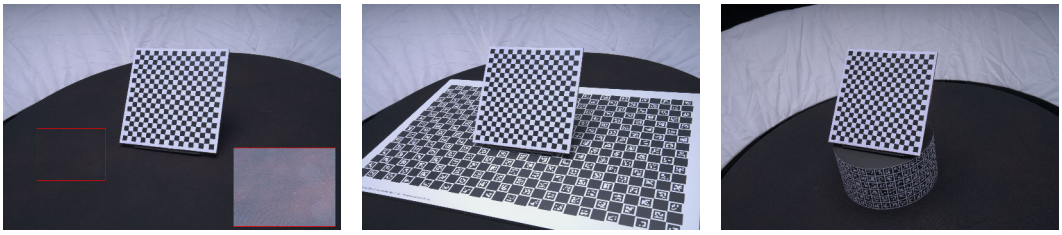
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A Camera Calibration Specifications

A.1 Calibration Accuracy

We compare our SfM-based camera calibration approach against three alternatives representative of existing benchmarks: COLMAP, a ChArUco board (similar to EvalMVX [21]), and an ArUco cylinder (similar to OpenSubstance [20]). As discussed in the main paper, accuracy is measured by the reprojection error on a separate test checkerboard not used during the calibration process. Our method achieves the highest precision with an error of 0.542 px (0.038 mm), followed by the ArUco cylinder (0.668 px), COLMAP (0.809 px), and ChArUco (1.086 px). Figure 8 illustrates these different calibration scenarios.



(a) Textured cloth (ours) (b) ChArUco board (c) ArUco cylinder

Figure 8: Calibration setup comparison. All configurations include a test checkerboard used only for evaluation. (a) Our approach uses SfM on the textured black cloth. The inset shows a brightness-enhanced crop of the cloth micro-texture, which provides point correspondences at the ultra-high image resolution. (b) ChArUco board [37]. (c) ArUco cylinder on the turntable base.

A.2 SfM Configuration Details

Camera calibration is performed with Meshroom [3] using DSP-SIFT [30] features, which improve robustness to scale changes compared to standard SIFT by pooling descriptors across multiple domain sizes. Meshroom is used with `ultra` preset and the SfM matching is configured in exhaustive mode with the following parameters: distance ratio $= 0.75$, maximum reprojection error $= 1.0$ px. Meshroom is run directly on the native 16-bit EXR images, preserving the full radiometric precision of the acquisition and avoiding the degradation of sub-pixel feature localisation that can result from 8-bit quantisation. The resulting bundle adjustment achieves a root-mean-square reprojection error of 0.095 px, used here as an internal consistency metric.

B Segmentation & Alignment Specifications

B.1 Canny Explanation

To establish accurate 2D–3D registration, we employ a contour-based alignment protocol that refines initial segmentation boundaries through optimized edge detection.

Contour extraction. For each view i, target contour points $\mathcal{C}_i$ are extracted from the single-light images captured under rotation-invariant ambient illumination. For highly specular objects, images acquired with sublimating opacifying spray are used instead to ensure clean silhouettes. A search band of $m = 20$ pixels is constructed around the initial Segment Anything Model (SAM) mask boundary by morphological dilation and erosion:

$$\mathcal{M}_i = \mathrm{dilate}(S_i, m) \setminus \mathrm{erode}(S_i, m), \tag{1}$$

and the target contour is defined as $\mathcal{C}_i = \mathrm{Canny}(I_i) \cap \mathcal{M}_i$, retaining only edge pixels that fall within this band.

Registration objective. The alignment minimizes the sum of nearest-neighbor distances between the projected silhouette vertices of the ground-truth mesh and the target Canny contours:

$$\min_{R,\,\mathbf{t},\,s} \sum_{i=1}^{N} \sum_{j \in \mathcal{S}(M,\,C_i)} d(\pi(R\,(s\,\mathbf{x}_j) + \mathbf{t},\; K_i)\,,\; \mathcal{C}_i)\,, \tag{2}$$

where $\mathbf{x}_j$ are vertices of the ground-truth mesh and $\mathcal{S}(M, C_i)$ is the set of silhouette vertices visible from camera C_i. The set $\mathcal{S}$ includes edges whose adjacent faces have opposite front/back visibility, filtered by rasterization to retain only occluding contour points. The perspective projection $\pi(\cdot, K_i)$ accounts for radial distortion, and $d(\cdot, \mathcal{C}_i)$ represents the nearest-neighbor distance to the target contour via a KD-tree. The solver uses the Trust Region Reflective method with a soft-ℓ_1 loss for robustness to spurious Canny edges.

Threshold optimization. The Canny edge detection thresholds are selected per object from a grid of 20 configurations: $\tau_\ell \in \{5, 10, 20, 50\}$ and $\tau_h \in \{20, 30, 40, 60, 80, 120\}$ with the constraint $\tau_\ell < \tau_h$, a Sobel kernel size of 15, and a search margin of 20 px. For each configuration, a full registration is performed and the result is evaluated by rasterizing the ground-truth mesh at the registered pose to produce a reference silhouette S for each view.

For a given tolerance ε (in pixels), the evaluation metrics are aggregated across views:

- **Silhouette coverage** $T_R = |\{s \in S : d(s, C) \leq \varepsilon\}| \,/\, |S|$: fraction of the GT silhouette covered by Canny edges.
- **Precision after refinement** $P_{\text{trim}} = |\{c \in C : d(c, S) \leq \varepsilon\}| \,/\, |C|$: fraction of refined Canny points that lie on the GT silhouette.
- $F_{0.5}$ **score**:

$$F_{0.5} = \frac{1.25 \cdot T_R \cdot P_{\text{trim}}}{0.25 \cdot P_{\text{trim}} + T_R}, \tag{3}$$

  which weights precision more than recall, favoring clean contours over high coverage.

The configuration maximizing $F_{0.5}$ at $\varepsilon = 2$ px is retained for each object.

B.2 Limitations of the IoU Score for Alignment Validation

EvalMVX [21] uses IoU between projected ground-truth masks and segmented image masks as its alignment-quality metric. Our analysis suggests that such global overlap scores can remain very high despite visible contour errors. On two objects from our dataset, *rock* and *warrior_head*, SAM masks reach IoU values above 0.99 against manually segmented reference masks. However, their contours differ by 4.80 pixels and 4.51 pixels on average, respectively, boundary shifts, which are poorly captured by IoU. See Figure 9.

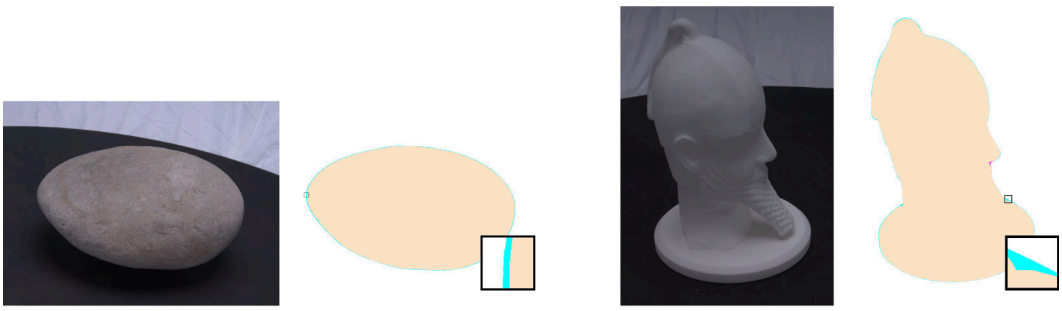


Figure 9: IoU limitations in assessing boundary precision. Visualization of the discrepancy between SAM (magenta) and Ground Truth (cyan) masks. The intersection (orange) represent the intersection of the masks. The boundary analysis reveals systematic contour shifts (averaging 4.80 to 4.51 px) that are critical for 2D-3D registration accuracy.

As shown in Table 3, incorporating Canny edge detection into our segmentation pipeline significantly enhances contour extraction compared to standard SAM masks. This refinement reduces the average contour distance from the 4–5 pixel range to between 1 and 2 pixels.

Table 3: **Contour refinement.** The proposed Canny-based refinement significantly reduces the mean contour distance (px).

	SAM		SAM + Canny (Ours)	
Object	**IoU ↑**	**Dist. (px) ↓**	**IoU ↑**	**Dist. (px) ↓**
Rock	0.9918	4.80	0.9972	1.28
Warrior Head	0.9913	4.51	0.9984	1.37

B.3 OpenSubstance Dataset Limitations

While the OpenSubstance [20] dataset provides a vast collection of objects, a significant portion of the library is unsuitable for rigorous qualitative or quantitative 3D reconstruction analysis. Figure 10 illustrates a representative subset of these problematic cases. The first two examples possess ground-truth models, yet their geometric characteristics pose significant challenges for COLMAP; this leads to substantial registration errors between the estimated camera poses and the ground-truth geometry. The two remaining examples have highly complex geometries that could not be captured reliably by the scanning process and therefore do not have ground-truth data. This information was shared by the dataset authors.



Figure 10: Sample of OpenSubstance objects that are unsuitable for rigorous 3D reconstruction analysis.

C Evaluation of the View-Invariant Lighting Setup

To validate the efficacy of our single-light acquisition regime, we conducted a two-stage experimental analysis focusing on the temporal stability and the rotational invariance of the single-light setup. As detailed in the main manuscript, this setup utilizes a top-mounted softbox flash and a reflective cone encircling the turntable. When perfectly aligned, this configuration is designed to produce a view-invariant lighting environment optimized for multi-view reconstruction.

Photometric stability. We first assessed the repeatability of the flash intensity by capturing 50 successive images of a static object from a fixed viewpoint. By computing the standard deviation for each pixel across the sequence, we observed an average standard deviation of 0.005 on a normalized intensity scale of $[0, 1]$. This low variance confirms the photometric stability of the flash and its suitability for consistent data acquisition.

Rotational invariance. To evaluate the view-invariant nature of the setup, we performed a symmetry analysis using a high-fidelity white sphere (25cm of diameter) positioned at the center of the turntable. Following camera calibration and the geometric alignment of a 3D spherical model with the image data, we identified a series of isophotes—contours of equal elevation—across the sphere's surface. Given the rotational symmetry of the capture environment, the radiance values along these isophotes should ideally remain constant regardless of the turntable's rotation. We extracted intensity data from 41 viewpoints across multiple isophotes spanning the height of the sphere. The resulting distribution, visualized in the box-and-whisker plots in Figure 11, demonstrates the precision of our lighting invariance. The observed standard deviations range between 0.005 and 0.025. Considering that a portion of this variance is attributable to the inherent flash instability identified previously, these low values confirm that the lighting remains largely independent of rotation.

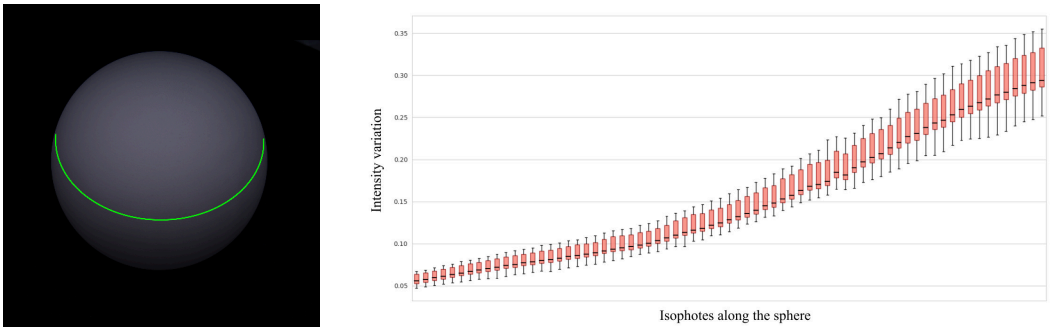


Figure 11: (Left) A visualization of a single isophote (green) on the calibration sphere. (Right) Distribution of pixel intensity variations for each isophote across the sphere's height. The standard deviation ranges from 0.0054 to 0.0267, with an average of 0.0141, confirming consistent illumination across viewpoints.

D Additional Calibration Files

D.1 Multi-light Calibration

Multi-light methods are evaluated on our dataset using normals predicted by UniMS-PS [25], a neural photometric-stereo method commonly used in recent multi-light reconstruction work and not requiring calibrated light sources. However, many calibrated photometric-stereo methods require light positions, directions, or intensities as input. To support such methods, we provide additional images for light calibration. Following standard sphere-based calibration procedures, we acquire multi-view multi-light sequences of two registered spheres placed in the object coordinate frame. The first is a Lambertian sphere of diameter 60.5 mm, that can be used to estimate photometric response and intensity-related quantities. The second is a specular sphere of diameter 60.2 mm, whose highlights provide constraints on light directions. Since the sphere poses are known from our image-to-geometry alignment, these sequences can be used to estimate the light parameters required by calibrated photometric-stereo models.

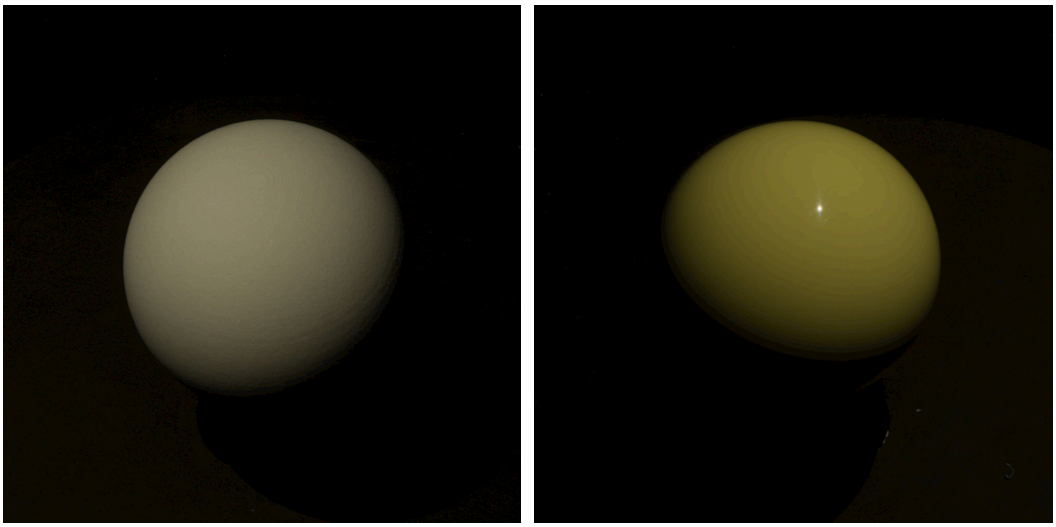


Figure 12: Light-calibration spheres. (Left) Lambertian sphere. (Right) Specular sphere.

D.2 Photometric Calibration

To support photometric accuracy and texture consistency, we include captures of a professional color calibration target (Calibrite ColorChecker Classic) for each LED and the single-light scenario. These

reference images allow for the correction of light color temperatures in photometric stereo pipelines and ensure that the final reconstructed meshes feature accurate and consistent textures.



Figure 13: Color calibration board seeing in one of the multi-light view (left), and with one LED turned on (right).

E Additional Information

E.1 Evaluation Scores

All quantitative evaluations are performed by comparing the reconstructed mesh $\mathcal{P}$ against the ground-truth mesh $\mathcal{G}$. Following standard benchmarking protocols, we report precision, recall, F-score, Chamfer distance, and Mean Angular Error (MAE).

The F-score is defined as the harmonic mean of precision $P(\epsilon)$ and recall $R(\epsilon)$ for a given distance threshold ϵ:

$$F(\epsilon) = \frac{2P(\epsilon)R(\epsilon)}{P(\epsilon) + R(\epsilon)}, \tag{4}$$

where precision and recall represent the fraction of points whose distance to the reference mesh is below ϵ:

$$P(\epsilon) = \frac{1}{|\mathcal{P}|} \sum_{\hat{\mathbf{x}} \in \mathcal{P}} [d_{\hat{\mathbf{x}} \to \mathcal{G}} < \epsilon] \quad \text{and} \quad R(\epsilon) = \frac{1}{|\mathcal{G}|} \sum_{\mathbf{x} \in \mathcal{G}} [d_{\mathbf{x} \to \mathcal{P}} < \epsilon]. \tag{5}$$

Here, $[\cdot]$ denotes the Iverson bracket.

To assess the proximity between the two surfaces, we define the distance from a reconstructed point $\hat{\mathbf{x}} \in \mathcal{P}$ to the ground-truth mesh $\mathcal{G}$ as:

$$d_{\hat{\mathbf{x}} \to \mathcal{G}} = \min_{\mathbf{x} \in \mathcal{G}} \|\hat{\mathbf{x}} - \mathbf{x}\|. \tag{6}$$

The distance from the ground truth to the reconstruction, $d_{\mathbf{x} \to \mathcal{P}}$, is defined analogously. These measurements are accumulated over the entire surface to compute the Chamfer Distance (CD):

$$CD = \frac{1}{2} \left(\frac{1}{|\mathcal{P}|} \sum_{\hat{\mathbf{x}} \in \mathcal{P}} d_{\hat{\mathbf{x}} \to \mathcal{G}} + \frac{1}{|\mathcal{G}|} \sum_{\mathbf{x} \in \mathcal{G}} d_{\mathbf{x} \to \mathcal{P}} \right). \tag{7}$$

To ensure the reliability of these metrics, it is essential that the point densities of the reconstructed mesh $\mathcal{P}$ and the ground-truth mesh $\mathcal{G}$ are comparable. A significant discrepancy in density can bias both the Chamfer Distance and the F-score, as sparsely populated regions may disproportionately affect the nearest-neighbor search. To mitigate this, we implement a consistent resampling pipeline applied to both the ground-truth and reconstructed meshes prior to evaluation. Each mesh is first converted into a dense point cloud via area-weighted upsampling, which adds points to each face in proportion to its surface area. This is followed by a spatial downsampling filter with a 0.05 mm radius to eliminate local clusters and enforce a uniform distribution. This procedure yields two point clouds with a constant inter-point distance of approximately 0.05 mm across all objects, providing a standardized and fair baseline for the final computation of the Chamfer Distance and F-score.

Curvature Estimation. To identify high-curvature regions, we perform a local geometric analysis for each vertex in the ground-truth mesh. Specifically, we define a spherical neighborhood within a radius of 2 mm around each vertex. Inside this neighborhood, the surface's principal curvatures are estimated, typically by fitting a local quadric surface to the tangent plane, to determine the maximum principal curvature. This local metric allows us to segment the mesh into regions of varying geometric complexity for a more detailed evaluation.

E.2 Hardware and Acquisition Parameters

The single lighting acquisition setup was controlled using a Godox DPIII1000V strobe with a maximum power of 1000 W, paired with a SmallRig RA-D120 softbox to ensure uniform light diffusion. All images were captured using a Sony α7RV camera. To maintain consistent depth of field across varying object geometries, a narrow aperture of $f/20$ and a sensitivity of ISO 100 were utilized for all captures.

The exposure settings were adjusted according to the acquisition paradigm as follows:

- **Single-light acquisition:** To effectively freeze the flash output and eliminate ambient light interference, the shutter speed was set to 1/250 s.
- **Multi-light acquisition:** To accommodate the LEDs intensities, a longer shutter speed of 1 s was employed.

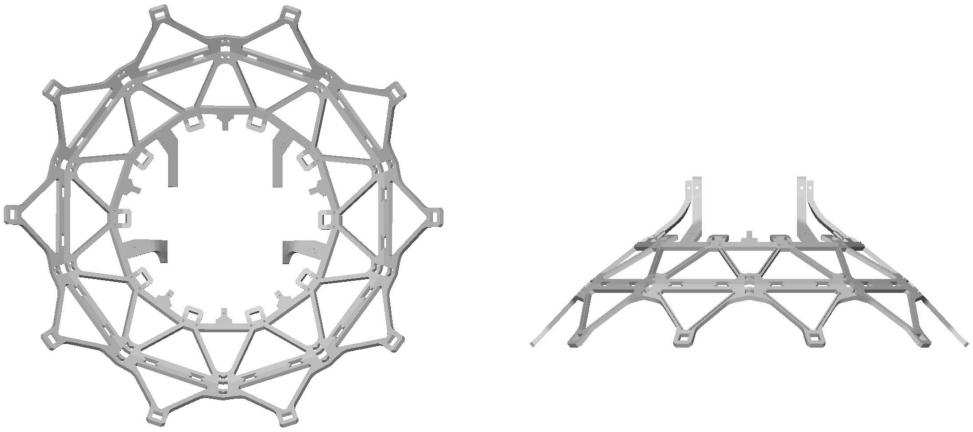


Figure 14: LED rig. Two views showing the spatial distribution of the LEDs around the camera.

E.3 Additional Results

Table 4 presents a detailed comparison of the Chamfer Distance (mm) across three image scales: d1, d2, and d4, corresponding to full, half, and quarter resolutions, respectively. Entries marked with a dash (-) indicate cases where a method did not yield a valid reconstruction at that specific resolution. This quantitative evaluation is conducted on a representative subset of the dataset, excluding four specific objects (plant, grid, bowl, and spine).

Table 4: Chamfer distance (mm) across methods and downscale levels. **Bold**: best per row, underline: second best. *n.r.* means not reported. *n.e.* means not evaluated.

	Meshroom			RealityScan			COLMAP			PGSR			NeuS2			Neuralangelo			RNb-NeuS2			SuperNormal		
Object	d1	d2	d4	d1	d2	d4	d1	d2	d4	d1	d2	d4	d1	d2	d4	d1	d2	d4	d1	d2	d4	d1	d2	d4
rock	0.107	0.099	0.109	0.131	0.121	0.168	*n.e.*	**0.074**	0.085	*n.e.*	2.434	2.412	*n.e.*	0.087	0.096	*n.e.*	1.066	1.745	*n.e.*	0.444	1.314	*n.e.*	*n.e.*	1.217
vase	0.163	0.224	0.418	**0.146**	0.197	0.578	*n.e.*	0.250	0.275	*n.e.*	1.965	1.980	*n.e.*	0.297	0.295	*n.e.*	0.492	0.519	*n.e.*	0.731	0.622	*n.e.*	1.086	1.010
hammer_head	0.107	0.101	0.119	2.254	0.108	0.173	*n.e.*	**0.063**	0.105	*n.e.*	1.025	1.040	*n.e.*	0.095	0.112	*n.e.*	0.794	1.049	*n.e.*	0.141	0.129	*n.e.*	*n.e.*	0.420
candle	1.133	1.322	1.492	0.807	1.153	1.438	*n.e.*	0.649	0.852	*n.e.*	1.940	1.906	*n.e.*	**0.273**	0.322	*n.e.*	0.466	0.477	*n.e.*	0.914	0.568	*n.e.*	1.832	0.682
can	0.159	0.179	0.328	**0.154**	0.180	0.263	*n.e.*	0.248	0.256	*n.e.*	2.297	2.296	*n.e.*	0.382	0.409	*n.e.*	0.650	0.628	*n.e.*	0.455	0.497	*n.e.*	2.698	1.848
glass	1.609	1.589	2.009	1.462	1.486	1.625	*n.e.*	1.317	1.378	*n.e.*	1.732	1.736	*n.e.*	0.933	1.054	*n.e.*	1.134	1.147	*n.e.*	1.343	1.341	*n.e.*	*n.e.*	*n.e.*
petanque_ball	0.167	0.149	0.203	0.158	0.209	0.355	*n.e.*	**0.142**	0.422	*n.e.*	1.995	1.997	*n.e.*	*n.e.*	0.268	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	0.175	*n.e.*	*n.e.*	0.454
jam_pot	1.457	1.550	1.401	0.981	0.676	1.086	*n.e.*	1.025	0.936	*n.e.*	2.073	2.086	*n.e.*	0.414	0.394	*n.e.*	1.154	1.209	*n.e.*	1.283	1.289	*n.e.*	*n.e.*	*n.e.*
colander	0.516	0.487	0.534	**0.431**	0.488	0.742	*n.e.*	0.620	0.545	*n.e.*	1.253	1.254	*n.e.*	*n.e.*	0.795	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*
wood	0.219	0.227	0.286	0.285	0.286	0.380	*n.e.*	**0.211**	0.271	*n.e.*	2.185	2.180	*n.e.*	*n.e.*	0.353	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	0.224	*n.e.*	*n.e.*	2.644
plate	*n.e.*	*n.e.*	*n.e.*	0.270	0.198	0.239	*n.e.*	*n.e.*	0.328	*n.e.*	*n.e.*	*n.e.*	*n.e.*	0.285	0.183	*n.e.*	0.651	0.467	*n.e.*	2.044	2.165	*n.e.*	*n.e.*	2.087
paw_pad	0.732	1.271	1.288	**0.368**	0.678	1.264	*n.e.*	0.383	0.694	*n.e.*	1.352	1.299	*n.e.*	0.430	0.503	*n.e.*	1.160	0.905	*n.e.*	1.020	1.040	*n.e.*	*n.e.*	2.591
moka_pot	0.167	**0.159**	0.215	0.171	0.240	0.310	*n.e.*	0.267	0.329	*n.e.*	1.703	2.141	*n.e.*	0.510	0.434	*n.e.*	1.105	1.058	*n.e.*	0.724	0.572	*n.e.*	1.086	0.771
lego	0.456	0.443	0.482	0.445	0.432	0.411	*n.e.*	0.493	0.452	*n.e.*	1.662	1.658	*n.e.*	**0.321**	0.324	*n.e.*	0.832	0.901	*n.e.*	0.709	0.722	*n.e.*	1.666	1.741
surgery_scissors	0.383	0.618	0.694	0.592	0.665	1.354	*n.e.*	0.289	0.477	*n.e.*	1.023	0.880	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	0.244	0.267	*n.e.*	*n.e.*	*n.e.*
knife	0.140	**0.092**	0.146	0.157	0.207	0.292	*n.e.*	0.097	0.114	*n.e.*	1.148	1.494	*n.e.*	*n.e.*	0.561	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	0.274	*n.e.*	*n.e.*	0.821
spikes_ball	0.585	0.578	0.792	0.465	0.557	0.763	*n.e.*	0.488	0.561	*n.e.*	1.612	1.593	*n.e.*	0.266	0.290	*n.e.*	0.780	0.936	*n.e.*	0.168	0.155	*n.e.*	*n.e.*	*n.e.*
straw_hat	0.276	0.222	0.260	0.231	0.239	0.278	*n.e.*	**0.198**	0.226	*n.e.*	1.832	1.848	*n.e.*	0.240	0.251	*n.e.*	1.417	1.343	*n.e.*	0.323	0.289	*n.e.*	*n.e.*	1.492
sneakers	0.355	0.327	0.322	0.293	**0.230**	0.274	*n.e.*	0.231	0.231	*n.e.*	1.537	1.505	*n.e.*	*n.e.*	0.284	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	0.390	*n.e.*	*n.e.*	1.297
surgery_saw	0.228	0.420	0.643	**0.221**	0.335	0.697	*n.e.*	0.379	0.505	*n.e.*	1.829	1.804	*n.e.*	*n.e.*	0.266	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	0.477	*n.e.*	*n.e.*	0.680
whisk	0.531	0.560	1.086	0.332	0.580	1.598	*n.e.*	0.356	0.423	*n.e.*	1.357	1.330	*n.e.*	0.448	0.915	*n.e.*	*n.e.*	*n.e.*	*n.e.*	**0.186**	0.213	*n.e.*	*n.e.*	1.845
warrior_head	**0.158**	0.186	0.271	0.210	0.206	0.449	*n.e.*	0.299	0.272	*n.e.*	1.789	1.774	*n.e.*	*n.e.*	1.008	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	0.798
plant	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*
grid	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	**0.841**	*n.e.*	*n.e.*	1.490
bowl	*n.e.*	*n.e.*	**1.105**	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	1.249	*n.e.*	*n.e.*	2.268
spine	*n.e.*	*n.e.*	0.822	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	*n.e.*	**0.578**	*n.e.*	*n.e.*	1.438
Mean	0.439	0.514	0.653	0.480	0.431	0.670	*n.e.*	0.385	0.443	*n.e.*	1.702	1.724	*n.e.*	**0.356**	0.434	*n.e.*	0.900	0.953	*n.e.*	0.715	0.669	*n.e.*	1.195	1.150

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